

Elmendorf's Theorem

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1 Introduction

Elmendorf's theorem is the statement that the category of G -spaces and the category of contravariant functors from the orbit category to $\mathbf{Top}$ have equivalent homotopy theories. That is, there is a Quillen equivalence between the model categories $\mathbf{GTop}$ and $\mathcal{O}_G\mathbf{Top}$. Our goal is to build model structures on $\mathbf{GTop}$ and $\mathcal{O}_G\mathbf{Top}$ such that Elmendorf's theorem is an immediate corollary of definitions. To do this, we first review the framework of model categories and Quillen equivalences while following the motivating example of the Quillen model structure on $\mathbf{Top}$. We lift this model structure to $\mathbf{GTop}$ and $\mathcal{O}_G\mathbf{Top}$ in such a way that generalises important homotopy theoretic ideas, and then we find that Elmendorf's theorem directly follows from this construction.

2 Model Categories

We begin by recalling the definition of a model category and realising the model category structures we have been working with in categories such as $\mathbf{Top}$ and $\mathbf{GTop}$. Model categories are used as a framework for certain lifting properties and factorisations of maps. That is, we define certain classes of morphisms in a category, and then introduce axioms about how these morphisms are related to each other by lifting and factorisation properties. This is useful because it allows us to extract the important properties of certain maps in homotopy theory, and generalise these ideas to, say, equivariant notions of homotopy theory. Further, given a notion of weak equivalence in a category, we would like to formally invert these weak equivalences so that we can study this category only up to homotopy. A model structure on a category allows us to do this with more control over the objects in the homotopy category. That is, we instead only take the objects that are both fibrant and cofibrant when we reduce to the homotopy category, and this turns out to capture the actual homotopy behaviour we care about.

Let $\mathcal{C}$ be a category. To define what a model structure on $\mathcal{C}$ is, we need some preliminary definitions. For the definitions in this section, we follow Rezk [1].

Definition 2.1. Given two morphisms $f : A \rightarrow B$, $g : X \rightarrow Y$, we say that f has the *left lifting property (LLP)* with respect to g (or g has the *right lifting property (RLP)* with respect to f) if for any commutative square,

$$\begin{array}{ccc} A & \longrightarrow & X \\ f \downarrow & & \downarrow g \\ B & \longrightarrow & Y \end{array}$$

there is a lift $s : B \rightarrow X$ such that the below diagram commutes.

$$\begin{array}{ccc} A & \xrightarrow{\quad} & X \\ f \downarrow & \nearrow \exists s & \downarrow g \\ B & \xrightarrow{\quad} & Y \end{array}$$

We write $f \boxtimes g$ to denote that f has the left lifting property with respect to g (or g the RLP with respect to f). If A is a set of morphisms in $\mathcal{C}$, then we write $f \boxtimes A$ if $f \boxtimes g$ for all $g \in A$. Similarly, we write $A \boxtimes g$ if $f \boxtimes g$ for all $f \in A$.

Note that these definitions are not asking for a unique lift, just the existence of a lift is required.

Definition 2.2. The *right complement* $A^\boxtimes$ of a set of morphisms A is

$$A^\boxtimes = \{g \mid a \boxtimes g \text{ for all } a \in A\}.$$

Similarly, the *left complement* ${}^\boxtimes A$ of a A is

$${}^\boxtimes A = \{f \mid f \boxtimes a \text{ for all } a \in A\}.$$

All of these definitions are designed to generalise the case $\mathcal{C} = \text{Top}$, so we will talk a familiar example in Top .

Example 2.3. Let $\mathcal{C} = \text{Top}$ and $g : E \rightarrow B$ a Serre fibration. That is, g is defined to be a map such that any commutative square belows has a lift.

$$\begin{array}{ccc} D^n \times \{0\} & \xrightarrow{\quad} & E \\ \downarrow & \nearrow \exists & \downarrow g \\ D^n \times I & \xrightarrow{\quad} & B \end{array}$$

If we let $A = \{D^n \times \{0\} \rightarrow D^n \times I\}$ be the set of inclusions $D^n \times \{0\} \rightarrow D^n \times I$, then g being a Serre fibration is exactly the statement that $g \in A^\boxtimes$.

Remark 2.4. Example 2.3 shows that the Quillen model structure on Top is *cofibrantly generated* [5]. We will briefly note when a model structure being cofibrantly generated is important, but we will not formally define what this means, nor will it be our focus.

A reasonable thing to ask of a class of maps is for it to be closed under composition. However, we will need a stronger condition for some of the morphisms we will deal with.

Definition 2.5. We say that a class of morphisms W satisfies the *2-out-of-3 property* if W contains all identities, and whenever 2 of f, g, gf are elements of W , so is the third.

Note that this definition is only asking about triples (f, g, gf) when the composition gf is defined.

Example 2.6. An easy example of a class of morphisms that satisfies 2-out-of-3 is simply the set of isomorphisms in any category. Further, in our favourite example of $\mathcal{C} = \text{Top}$, we have that the weak equivalences satisfy 2-out-of-3. This is because if two of f, g, gf are weak equivalences, then two of f_*, g_*, g_*f_* are isomorphisms. Therefore all three are isomorphisms and hence all three of f, g, gf are weak equivalences.

We have discussed lifting properties and composition properties of our classes of morphisms, so now we discuss factorisation properties.

Definition 2.7. A *weak factorisation system* is a pair of classes of maps (L, R) such that

1. Every morphism f factors as $f = rl$ with $l \in L, r \in R$.
2. $L = {}^\square R, R = L^\square$

Example 2.8. let TFib be the class of Serre fibrations in Top that are also weak equivalences and let $\text{Cof} = {}^\square(\text{TFib})$. The pair $(\text{Cof}, \text{TFib})$ is a weak factorisation system of Top . We will not prove this, but we note that the class Cof is the saturated class of boundary inclusions $S^{n-1} \rightarrow D^n$ [6].

Finally, we can define a model category.

Definition 2.9. A *model category* M is a category with three distinguished classes of maps W , Cof , Fib such that

1. M has all (small) limits and colimits
2. W obeys 2-out-of-3
3. $(\text{TCof}, \text{Fib})$ and $(\text{Cof}, \text{TFib})$ are weak factorisation systems where $\text{TCof} = \text{Cof} \cap W$ and $\text{TFib} = \text{Fib} \cap W$.

The classes $W, \text{Fib}, \text{Cof}$ are called the weak equivalences, fibrations and cofibrations respectively. The classes TFib, TCof are called the trivial fibrations and the trivial cofibrations respectively.

Remark 2.10. We can see from the requirement that $(\text{TCof}, \text{Fib})$ and $(\text{Cof}, \text{TFib})$ are weak factorisation systems that any two classes $W, \text{Cof}, \text{Fib}$ completely determines the third. So when defining a model category, we will only ever define two of the three classes, and there is no ambiguity about what the third class must be.

In this definition, it looks like we are only distinguishing some classes of maps. However, this also gives us a distinction of certain objects.

Definition 2.11. An object $X \in \mathcal{C}$ is *fibrant* if the map to the terminal object $X \rightarrow *$ is a fibration. Dually, X is *cofibrant* if the map from the initial object $\emptyset \rightarrow X$ is a cofibration.

Example 2.12. The notation used in Example 2.8 seems to suggest that $\mathbf{Top}$ is a model category. This is true, and there are multiple model structures that can be put on $\mathbf{Top}$. The two standard choices are the Quillen model structure and the Hurewicz model structure, the main difference between them being the choice of weak equivalences. In the Quillen model, the weak equivalences are the weak homotopy equivalences and the fibrations are the Serre fibrations. In the Hurewicz model, the weak equivalences are the homotopy equivalences and the fibrations are the Hurewicz fibrations. The Quillen model has been our main motivating example through this whole section, and it will continue to be our focus and concrete example. In the Quillen model, every object is fibrant because the map $X \rightarrow *$ is a Serre fibration, and the cofibrant objects are the CW complexes [6].

3 Homotopy Category and Quillen Equivalence

Now that we understand the notion of a model on a category, we construct the homotopy category $\mathbf{Ho}(\mathcal{C})$. That is, we construct the category where we only care about objects and morphisms up to homotopy.

Definition 3.1. The *homotopy category* $\mathbf{Ho}(\mathcal{C})$ of a model category $\mathcal{C}$ is the category with objects the objects in $\mathcal{C}$ which are both fibrant and cofibrant, and morphisms homotopy classes of morphisms in $\mathcal{C}$.

The definition of homotopy class agrees with our definition in $\mathbf{Top}$. The generalisation of this definition to an arbitrary model category is given in [9]. The idea is to use the factorisation given from the model structure to construct a good notion of a cylinder object so that a homotopy can be a map from $\mathbf{Cyl}(X) \rightarrow Y$ where $\mathbf{Cyl}(X) = X \times I$ in the case of CW complexes. The upshot is a model on a category can be used to formally work with our category only up to homotopy.

Having defined model categories, we want a notion of equivalence of model categories (and hence, an equivalence of homotopy categories). This is called a Quillen equivalence, which we define after proving a small lemma.

Lemma 3.2. Let $F : \mathcal{M} \rightleftarrows \mathcal{N} : G$ be an adjunction between model categories. Then both of the following hold.

1. F preserves $\mathbf{TCof}$ if and only if G preserves $\mathbf{Fib}$
2. F preserves $\mathbf{Cof}$ if and only if G preserves $\mathbf{TFib}$

Proof. Suppose that G preserves fibrations and that f is a trivial cofibration. We want to show that Ff is a trivial cofibration. This is the same as showing that $Ff \in {}^{\mathbb{Q}}(\text{Fib})$, because $(\text{TCof}, \text{Fib})$ is a weak factorisation system. So suppose $g \in \text{Fib}$. Then we need to find a lift in the below square.

$$\begin{array}{ccc} FA & \longrightarrow & X \\ Ff \downarrow & \nearrow & \downarrow g \\ FB & \longrightarrow & Y \end{array}$$

Applying G to the square we obtain an equivalent lifting problem.

$$\begin{array}{ccccc} A & \xrightarrow[\sim]{\eta_A} & GFA & \longrightarrow & GX \\ f \downarrow & & \downarrow GFf & \nearrow & \downarrow Gg \\ B & \xrightarrow[\sim]{\eta_B} & GFB & \longrightarrow & GY \end{array}$$

Here η is the counit of the adjunction. Since G preserves fibrations, Gg is a fibration and thus we have such a lift. Applying F and composing with the unit, we obtain a lift for the original lifting problem. Thus F preserves trivial cofibrations. The other direction and point (2) follows from a completely identical argument. $\square$

We want our notion of equivalent model categories to be some sort of adjunction that preserve the classes of morphisms. So we make the following definition.

Definition 3.3. An adjunction $F : M \rightleftarrows N : G$ is a *Quillen pair* if F preserves cofibrations and trivial cofibrations.

Note that Lemma 3.2 says that this is equivalent to G preserving fibrations and trivial fibrations, so this definition is not arbitrarily asymmetric. Further, a Quillen equivalence sends fibrant objects to fibrant objects, cofibrant objects to cofibrant objects and homotopies to homotopies. So a Quillen equivalence between model categories gives an equivalence of their homotopy categories. Thus, our goal will be to define model structures on GTop , $\mathcal{O}_G\text{Top}$ and prove that these structures are Quillen equivalent, showing that they have equivalent homotopy categories.

4 Model Structures on GTop and $\mathcal{O}_G\text{Top}$

Now we go equivariant with the goal of studying the homotopy theory of spaces equipped with a G -action. The first question is what the correct notion of equivariant homotopy theory is. That is, what model structure do we put on GTop . For this section, we are loosely following [2] and [4].

Firstly, we want think about G actions on spaces in a way that can be easily generalised to objects in other categories. To do this, we now let G be a one object category, instead of a finite group. The morphisms in this category are all automorphisms, and they each correspond to the action of an element of (the group) G on itself. When thought of in this way, we find that the data of a group action on a space X is exactly the data of a functor $F : G \rightarrow \text{Top}$ such that the object of G is sent to X . Further, a natural transformation between two such functors is exactly the data of a G -equivariant map. Thus, we think of $\text{GTop} = \text{Fun}(G, \text{Top})$, the functor category from G to Top . This generalisation is useful because Top can be replaced with a different category $\mathcal{C}$ and under some assumptions about $\mathcal{C}$, everything following will work the same way.

Defining GTop in this way, we notice that GTop inherits a seemingly natural model structure from Top . We said above that a morphism in GTop can be realised as a G -equivariant map in Top . So we can define a morphism in GTop to be a weak equivalence or fibration if it is respectively a weak equivalence or fibration in Top (Definition 4.2 and Remark 4.3 gives that this is a model structure on GTop). However, this definition does not capture the homotopy theory we want to generalise. One reason is that this model structure does not give an equivariant Whitehead theorem. Instead, we take a different approach [2].

When thinking of G as a group and $H \subset G$ a subgroup of G , we have a fixed point functor $(-)^H : \text{GTop} \rightarrow \text{Top}$ that maps a G -space X to X^H , the H -fixed points of X . When thinking of G as a one-object category, we can think of H as a subcategory of G . Then a functor $X : G \rightarrow \text{Top}$ restricts to a diagram $X|_H : H \rightarrow \text{Top}$ in Top . So instead we define X^H to be the limit of the diagram $X|_H$. We notice that this definition agrees with the fixed point functor definition, however this definition does not rely on elements. That is, this definition still makes sense when Top is replaced with an arbitrary category $\mathcal{C}$, so long as $\mathcal{C}$ has small limits.

Consider the orbit category $\mathcal{O}_G$. In a similar way to defining GTop , we define $\mathcal{O}_G\text{Top} = \text{Fun}(\mathcal{O}_G^{\text{OP}}, \text{Top})$. Now for any $X \in \text{GTop}$, we can define a functor

$$\begin{aligned}\Phi(X) : \mathcal{O}_G^{\text{OP}} &\rightarrow \text{Top} \\ \Phi(X)(G/H) &= X^H.\end{aligned}$$

If we have a map $G/H \rightarrow G/K$, then H is a subcategory of K . So the limit over the diagram $X|_K$ is a cone on $X|_H$. Therefore we obtain a unique map $X^K \rightarrow X^H$ by the universal property of limits. This is how $\Phi(X)$ acts on morphisms, and we see that this makes $\Phi(X)$ contravariant. We have defined a functor $\Phi(X)$ for every $X \in \text{GTop}$, so we have a functor $\Phi : \text{GTop} \rightarrow \mathcal{O}_G\text{Top}$.

Consider the functor $\Lambda : \mathcal{O}_G\text{Top} \rightarrow \text{GTop}$ defined by

$$\Lambda(Y) = Y(G/e)$$

The full subcategory on G/e has automorphisms that correspond to automorphisms of G , so it is naturally isomorphic to the category G . That is, $Y(G/e)$ inherits a G -action from the automorphism of G/e . Hence Λ is GTop valued. Now we prove that Λ is the left adjoint of Φ .

Proposition 4.1. We have an adjunction $\Lambda : \mathcal{O}_G\text{Top} \rightleftarrows \text{GTop} : \Phi$.

Proof. This is a matter of unpacking definitions. We want to show that

$$\mathrm{Hom}_{\mathbf{GTop}}(\Lambda Y, X) \cong \mathrm{Hom}_{\mathcal{O}_G \mathbf{Top}}(Y, \Phi(X))$$

where $Y \in \mathcal{O}_G \mathbf{Top}$ and $X \in \mathbf{GTop}$. That is, we want to show that

$$\mathrm{Hom}_{\mathbf{GTop}}(Y(G/e), X) \cong \mathrm{Hom}_{\mathcal{O}_G \mathbf{Top}}(Y, (G/H \mapsto X^H)).$$

A natural transformation $f \in \mathrm{Hom}_{\mathbf{GTop}}(Y(G/e), X)$ is exactly the data of a square below which commutes for all $g \in G$.

$$\begin{array}{ccc} Y(G/e) & \xrightarrow{g} & Y(G/e) \\ f \downarrow & & \downarrow f \\ X & \xrightarrow{g} & X \end{array}$$

A natural transformation $g \in \mathrm{Hom}_{\mathcal{O}_G \mathbf{Top}}(Y, (G/H \mapsto X^H))$ is completely determined by its component on G/e . That is, it is completely determined by a G -equivariant map $g_{G/e} : Y(G/e) \rightarrow X$. This is because for any map $G/e \rightarrow G/H$, the induced map $X^H \rightarrow X$ is unique by our limit definition (in $\mathbf{Top}$, it is just an inclusion of fixed point spaces). So the G -action on X^H is the restriction of the G -action on X and thus the only choice of data is $g_{G/e}$. We see that this is the same data as the commutative square above, so we have a bijection which is certainly natural. That is, we have shown that Λ is the left adjoint to Φ . $\square$

While lifting a model structure from $\mathbf{Top}$ onto $\mathbf{GTop}$ came with issues, we lift our model structure on $\mathbf{Top}$ to $\mathcal{O}_G \mathbf{Top}$ in the same way. The difference is lifting it to $\mathcal{O}_G \mathbf{Top}$ actually captures the homotopy theoretic results we want.

Definition 4.2 (Projective Model Structure). A morphism of $f \in \mathrm{Hom}_{\mathcal{O}_G \mathbf{Top}}(X, Y)$ is a *projective fibration* (resp. *projective weak equivalence*) if for all $H \subset G$

$$f_{G/H} : X(G/H) \rightarrow Y(G/H)$$

is a fibration (resp. weak equivalence) in $\mathbf{Top}$. This defines a model structure, known as the projective model structure.

Remark 4.3. This definition can also be made for $\mathbf{GTop}$ and in fact any functor category $\mathrm{Fun}(G, \mathcal{C})$ where $\mathcal{C}$ is a model category. However the proof that this is a model structure relies on $\mathcal{C}$ being cofibrantly generated.

We want to show that the projective model structure on $\mathcal{O}_G \mathbf{Top}$ (and $\mathbf{GTop}$) is actually a model structure. Limits in $\mathrm{Fun}(G, \mathcal{C})$ are computed object-wise, so $\mathrm{Fun}(G, \mathcal{C})$ certainly contains all small limits and colimits. By repeating the proof in Example 2.6, we see that the projective weak equivalences defined this way obey 2 out of 3. The main obstruction to proving this is showing the weak factorisation system requirements.

Lemma 4.4 (Quillen's Factorisation Lemma). Any morphism $f : X \rightarrow Y$ of $\mathcal{O}_G\text{Top}$ (resp. GTop) may be factored $f = pg$ where g is a projective cofibration and p is a trivial projective fibration.

This lemma fills in the holes and proves that this projective model structure is actually a model structure. However, the proof of this is very long winded and relies heavily on the small object argument. To avoid the lengthy details, we omit this proof, noting that the key features of it are the small object argument and the fact that the model structure on Top is cofibrantly generated. That is, the whole construction of the factorisation uses the generating set of the cofibrations, and then applies the small object argument to this construction. It is also shown in this construction that this projective model structure is cofibrantly generated [5].

Now we have a model structure on $\mathcal{O}_G\text{Top}$. We also have a functor $\Phi : \text{GTop} \rightarrow \mathcal{O}_G\text{Top}$, and the model structure on $\mathcal{O}_G\text{Top}$ is cofibrantly generated. Therefore, applying the same technique in Quillen's factorisation lemma, we can lift the model structure on $\mathcal{O}_G\text{Top}$ to GTop . That is, $f : X \rightarrow Y$ in GTop is a fibration (resp. weak equivalence) if $f^H : X^H \rightarrow Y^H$ is a fibration (resp. weak equivalence). We notice that this agrees with our original definition of an equivariant weak equivalence. Using this as our model, we obtain an equivariant version of Whitehead's theorem and have captured the original homotopy theory ideas that we were trying to generalise.

After putting all the hard work into making the correct definitions, Elmendorf's theorem follows simply from the fact that Φ is defined to preserve fibrations and trivial fibrations.

Theorem 4.5 (Elmendorf). The adjunction $\Lambda : \mathcal{O}_G\text{Top} \rightleftarrows \text{GTop} : \Phi$ is a Quillen equivalence.

That is, we have shown that the homotopy category of $\mathcal{O}_G\text{Top}$ is equivalent to the homotopy category of GTop . Further, the model category structures on these categories are the natural choices because they generalise the important theorems from ordinary homotopy theory. In summary, we have described the important and familiar model structures on Top , GTop and $\mathcal{O}_G\text{Top}$, and shown that under a good choice of model for these categories, we get Elmendorf's theorem essentially for free.

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